Covariant Biquaternionic Geometry of UBT

Connections, Torsion, Rank, and Integrability for Students

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Chapter 1

Covariant Derivatives, Connections, and the Clean UBT Geometry

This chapter explains the geometric core of UBT in the form closest to the original intuition of the theory:

$$\Theta \quad \longrightarrow \quad E_\mu := D_\mu\Theta \quad \longrightarrow \quad \{E_\mu, E_\nu\} \text{ and } [D_\mu, D_\nu].$$

The anticommutator carries the metric, while the commutator of covariant derivatives carries curvature. No embedding map, compact-fiber average, or observer projection is needed in the minimal local construction.

1.1 Ordinary, partial, and covariant derivatives

For an ordinary scalar function $f(x)$, the derivative df/dx measures how the number f changes when x changes. For a field on spacetime we use partial derivatives,

$$\partial_\mu\Theta := \frac{\partial\Theta}{\partial x^\mu}, \qquad \mu = 0, 1, 2, 3.$$

A partial derivative compares components written in a fixed basis. This is sufficient if the same basis can be used everywhere. Geometry becomes subtler when the local frame itself rotates or boosts from point to point.

A simple analogy is a vector written in a rotating coordinate frame. Even a physically constant vector acquires changing components because the basis is changing. The covariant derivative corrects for this basis change:

$$D_\mu\Theta = \partial_\mu\Theta + \rho_*(\Omega_\mu)\Theta.$$

Here Ω_μ is the connection and ρ_* tells us how the connection acts in the representation of Θ. For biquaternions, a left action, a right action, and a two-sided action are genuinely different and must not be silently interchanged.

1.2 What is a connection?

A connection is not a force by itself. It is a rule for comparing local frames at neighboring points. The symbol Ω_μ is the UBT internal or biquaternionic-frame connection.

Christoffel symbols,

$$\Gamma^\rho{}_{\mu\nu},$$

are closely related, but they act on coordinate indices. For example,

$$\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma^\rho{}_{\mu\sigma}V^\sigma.$$

Thus:

- ${\Gamma^\rho}_{\mu\nu}$ tells us how the coordinate basis changes;
- ${\omega_\mu}^a{}_b$, and its spin/biquaternionic lift Ω_μ, tell us how the local Lorentz frame changes.

They are related by the tetrad compatibility identity

$$\partial_\mu {e_\nu}^a - {\Gamma^\rho}_{\mu\nu} {e_\rho}^a + {\omega_\mu}^a{}_b {e_\nu}^b = 0.$$

They are therefore two descriptions of the same geometric transport, not the same coefficients. In particular, $\Gamma = \operatorname{Re}\Omega$ is not a valid identification.

1.3 Why the classical connection is not an extra free field

For a nondegenerate tetrad, metric compatibility and zero torsion determine a unique Lorentz connection:

$$\boxed{\mathring{\omega}_\mu{}^a{}_b = e_b{}^\nu \left(\mathring{\Gamma}^\rho{}_{\mu\nu}(g) e_\rho{}^a - \partial_\mu e_\nu{}^a \right).}$$

The circle marks the torsion-free Levi–Civita connection. Thus, in the ordinary torsion-free GR branch, the frame connection is computed from the tetrad just as the Christoffel symbols are computed from the metric. A local Lorentz rotation changes the numerical coefficients of the tetrad and the connection but does not change the geometry.

1.3.1 Torsion and contorsion

The torsion two-form is

$$T^a = de^a + {\omega^a}_b \wedge e^b.$$

A general metric-compatible connection can be written

$$\boxed{\omega = \mathring{\omega}(e) + K(T),}$$

where K is the contorsion. With $T^a = \frac{1}{2} {T^a}_{bc} e^b \wedge e^c$ and the convention above,

$$\boxed{K_{abc} = \frac{1}{2} \left(T_{cab} - T_{abc} - T_{bca} \right).}$$

This formula has an important conceptual consequence: once the tetrad and the torsion are specified, the metric-compatible connection is unique. The open full-UBT question is therefore not an arbitrary leftover Ω_μ; it is the dynamical law selecting T.

If the UBT vacuum equations imply $T = 0$, the connection reduces to the Levi–Civita spin connection. If spin or another biquaternionic source creates torsion, the same theorem reconstructs the connection from that torsion.

1.4 Flat space and an explicit UBT representer

In flat Minkowski spacetime, using inertial Cartesian coordinates and a constant local frame, one can choose

$$\Gamma^\rho{}_{\mu\nu} = 0, \qquad \Omega_\mu = 0, \qquad D_\mu = \partial_\mu.$$

This statement can be strengthened from a choice of gauge to an explicit single-field representer. Let

$$\boxed{\Theta_{\mathrm{SR}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0} \left(i x^0 \mathbf{1} + x^k \mathbf{e}_k \right).}$$

Then

$$\mathcal{N}_0^{-1/2} \partial_0 \Theta_{\mathrm{SR}} = i\mathbf{1}, \qquad \mathcal{N}_0^{-1/2} \partial_k \Theta_{\mathrm{SR}} = \mathbf{e}_k.$$

The central anticommutator of these four constant directions is the Minkowski metric. Every second spacetime derivative of Θ_{SR} vanishes, so it also solves the standard source-free constant-coefficient second-order master operator in this flat branch.

More generally, for any constant Lorentz tetrad $\bar{E}_\mu$,

$$\Theta_{\mathrm{aff}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0}\,\bar{E}_\mu x^\mu$$

generates that tetrad exactly. This closes the special-relativistic representer problem, although it does not prove uniqueness or curved-space existence.

In polar coordinates, or in an accelerating frame, Γ and Ω may be nonzero even though spacetime is flat. The invariant statement of flatness is vanishing curvature, not vanishing connection coefficients in every frame.

1.5 The four covariant gradients as a tetrad

Define

$$E_\mu := \frac{1}{\sqrt{\mathcal{N}_0}} D_\mu \Theta.$$

The four objects E_0, E_1, E_2, E_3 are the local UBT rulers and clock direction. They are not four extra matter fields; they are the first covariant derivatives of the single fundamental field Θ.

Let i be the ordinary commuting complex unit and $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ the quaternion units. In the classical Lorentz sector write

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k, \qquad e_\mu{}^a \in \mathbb{R}.$$

Quaternion conjugation changes the signs of the three quaternion units but leaves the complex scalar coefficient untouched:

$$E_\mu^\sharp = i e_\mu{}^0 \mathbf{1} - e_\mu{}^k \mathbf{e}_k.$$

1.6 Why the anticommutator is the metric

Quaternion multiplication contains a scalar product and a cross product. In the symmetrised product the cross products cancel:

$$\boxed{\frac{1}{2}\left(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu\right) = g_{\mu\nu}\mathbf{1}.}$$

The right-hand side contains $\mathbf{1}$, the unit biquaternion. The entire left-hand side is already central: an ordinary real number multiplied by the algebra unit. No trace, real-part map, phase projection, or fiber average is required.

Expanding the coefficients gives

$$g_{\mu\nu} = -e_\mu{}^0 e_\nu{}^0 + e_\mu{}^1 e_\nu{}^1 + e_\mu{}^2 e_\nu{}^2 + e_\mu{}^3 e_\nu{}^3,$$

which is exactly

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1, 1, 1, 1).$$

The minus sign of time comes from $i^2 = -1$.

1.7 What does the other half of the product do?

The antisymmetric algebraic part is

$$\Sigma_{\mu\nu} = \frac{1}{2}\left(E^\sharp_\mu E_\nu - E^\sharp_\nu E_\mu\right).$$

It describes an oriented plane or bivector and is a natural carrier for local rotation, boost, and spin information. It is not itself the curvature.

Three expressions must be kept separate:

$$\begin{aligned}
\{E_\mu, E_\nu\}_\sharp &\longrightarrow \text{metric},\\
[E_\mu, E_\nu]_\sharp &\longrightarrow \text{algebraic bivector/spin structure},\\
[D_\mu, D_\nu] &\longrightarrow \text{connection curvature or gauge field strength}.
\end{aligned}$$

1.8 Curvature from noncommuting derivatives

For a one-sided connection, the curvature has the familiar form

$$\mathcal{R}_{\mu\nu} = \partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu].$$

The final commutator is essential because biquaternion multiplication is noncommutative. Geometrically, curvature measures how a frame changes after transport around a small closed loop.

The equation $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$ creates an additional integrability condition. This is the point at which the representation of the connection becomes decisive.

1.9 Why the old rank problem disappears locally

A symmetric 4×4 metric has ten independent components. Looking only at one value of $\Theta \in \mathbb{B} \simeq \mathbb{R}^8$ suggests the misleading comparison $8 < 10$.

The metric is not built from the value of Θ alone. It is built from four covariant derivatives E_μ. In the Lorentz sector these form a real 4×4 tetrad ${e_\mu}^a$ with sixteen components. Six changes are local rotations and boosts that leave the metric unchanged. Therefore

$$16 - 6 = 10.$$

More rigorously, every small symmetric metric variation $h_{\mu\nu}$ is produced by

$$\delta {e_\mu}^a = \frac{1}{2} h_{\mu\rho} e^{\rho a}.$$

The tetrad-to-metric map therefore has rank ten at every nondegenerate tetrad. This resolves the local kinematic rank question. It does not yet prove that every curved tetrad is generated by an on-shell UBT field.

1.9.1 A correct calculation can answer the wrong architectural question

The former single-section rank obstruction was mathematically correct in an induced-metric framework where Einstein equations were to be recovered only from normal variations of one embedding-like section. It did not constitute a no-go theorem for the covariant tetrad $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$. The compact-fiber construction repaired the former formulation by adding profile directions, but the tetrad formulation shows that this repair was not required by the original architecture.

This distinction motivates a general research rule:

> Before adding fields, modes, dimensions, projections, averages, or auxiliary structures to solve an obstruction, first test whether the obstruction is an artefact of the formulation in which it was derived.

The fiber route remains mathematically consistent in its own framework. Its problem is weak selection and representer redundancy, not an algebraic contradiction.

1.10 The integrability problem

Once the classical connection is reconstructed from the tetrad and torsion, the defining equation becomes schematically

$$E_\mu = \mathcal{N}_0^{-1/2} \left(\partial_\mu \Theta + \rho_*(\Omega_\mu[E, T])\Theta\right).$$

The unknown E occurs on both sides. This is an implicit nonlinear first-order PDE or fixed-point system. Eliminating Ω kinematically does not by itself prove existence or uniqueness of a pair (Θ, E).

1.10.1 A one-sided flatness obstruction

Suppose the connection acts only from the left,

$$D^L_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta,$$

so that

$$[D^L_\mu, D^L_\nu]\Theta = F^L_{\mu\nu}\Theta.$$

In a torsion-free compatible tetrad branch, antisymmetrising the covariant Hessian gives

$$F^L_{\mu\nu}\Theta = 0.$$

If Θ is invertible as a biquaternion, then

$$F^L_{\mu\nu} = 0.$$

Thus the naive one-sided regular representation cannot generically support curved GR while retaining an invertible Θ and torsion-free tetrad compatibility. Noninvertible stabilizer configurations may evade this result, but they cannot be assumed to represent arbitrary geometries.

1.10.2 The natural two-sided biquaternionic derivative

Biquaternions naturally admit both left and right multiplication. Consider

$$\boxed{D_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta - \Theta B_\mu.}$$

A direct expansion gives

$$\boxed{[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}.}$$

Torsion-free compatibility therefore requires

$$F^A_{\mu\nu}\Theta = \Theta F^B_{\mu\nu}.$$

For invertible Θ,

$$\boxed{F^A_{\mu\nu} = \Theta F^B_{\mu\nu}\Theta^{-1}.}$$

Unlike the one-sided case, neither curvature must vanish. The field Θ intertwines the left and right curvature representations. This is a genuine structural narrowing of GAP-10I: the two-sided/bimodule derivative is the minimal algebra-native route that avoids the generic flatness obstruction.

It is not yet a complete curved-space solution. The action must still fix the relation between A_μ and B_μ, their involution properties, torsion, and boundary conditions.

1.11 Conditional dynamical subclosures

1.11.1 The algebraic Cartan torsion equation

In the minimal first-order Hilbert–Palatini branch, independent variation of the Lorentz connection gives

$$\epsilon_{abcd}T^a \wedge e^b = \kappa\tau_{cd}.$$

For a nondegenerate tetrad this is a 24×24 pointwise linear map of rank 24. Thus zero spin current gives $T = 0$, while a specified spin current gives a unique torsion and contorsion. This closes the torsion algebra inside the minimal branch, not the derivation of that branch from fundamental UBT.

1.11.2 Fixed-set propagation of the Lorentz slice

The anti-linear involution

$$\mathcal{J}X = -\overline{X^\sharp}$$

has the physical Lorentz module as its fixed set. If a well-posed unique Cauchy evolution is $\mathcal{J}$-equivariant and its data and sources are fixed by $\mathcal{J}$, uniqueness implies that the solution remains in the Lorentz slice. The final UBT action must still be checked against these hypotheses.

1.11.3 Imaginary-time metric stability

A vertical flow

$$\partial_\psi e_\mu{}^a = \Lambda_\psi{}^a{}_b e_\mu{}^b, \qquad \Lambda_{\psi ab} = -\Lambda_{\psi ba},$$

is tangent to a local Lorentz gauge orbit and satisfies $\partial_\psi g_{\mu\nu} = 0$ identically. A second sufficient mechanism is a unique ψ-translation-invariant evolution with ψ-independent initial data. Selecting the physical mechanism and excluding unstable non-gauge modes remains a dynamical task.

1.11.4 Augmented holonomy for prescribed curved coefficients

For prescribed E_μ, A_μ, B_μ, the inhomogeneous system

$$\partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu = \sqrt{\mathcal{N}_0}E_\mu$$

becomes homogeneous parallel transport for $Y = (\Theta, 1)^T$ in an augmented bundle. A single-valued solution with initial value Y_0 exists exactly when the augmented holonomy fixes Y_0; flat augmented curvature is sufficient on a simply connected domain. This closes prescribed-coefficient integrability, not self-consistent UBT selection of the coefficients.

1.11.5 The conditional Einstein–Lambda endpoint

The minimal Palatini branch with zero spin current yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Under the ordinary four-dimensional Lovelock assumptions, every local natural divergence-free second-order metric equation without extra light geometric fields has gravitational part $aG_{\mu\nu} + bg_{\mu\nu}$. Therefore the conditional low-energy endpoint is unique. UBT must still derive those assumptions, coefficients and matter terms from its canonical action.

1.12 Implicit versus transcendental equations

The words *implicit* and *transcendental* describe different properties.

An equation is implicit when the unknown occurs inside the map that is supposed to determine it:

$$E = \mathcal{F}(E, \Theta).$$

An equation is transcendental when it contains nonalgebraic functions such as an exponential, logarithm, trigonometric function, or Jacobi theta function:

$$E = e^{-E}.$$

The UBT tetrad equation is already implicit and differential. If the allowed field $\Theta(q,\tau)$ is restricted to a Jacobi-theta or another transcendental function class, then the complete system may be both implicit and transcendental. In a formal paper these properties should be stated separately, even though intuitively both make the unknown "entangled with itself."

1.13 Derivatives versus variations

A derivative such as $D_\mu\Theta$ describes how one field configuration changes through spacetime. A variation compares nearby possible configurations:

$$\Theta_\varepsilon = \Theta + \varepsilon\,\delta\Theta.$$

The variation is

$$\delta\Theta = \left.\frac{d\Theta_\varepsilon}{d\varepsilon}\right|_0.$$

Variations are needed when deriving field equations from an action. They are not replacements for spacetime derivatives. When the connection depends on the field,

$$\delta(D_\mu\Theta) = D_\mu(\delta\Theta) + \rho_*(\delta\Omega_\mu)\Theta$$

and the induced connection variation must be included.

1.14 What is closed and what remains open

The following results are closed at their stated levels:

1. the central anticommutator metric and rank-ten tetrad map;
2. GAP-10Ω-KIN/GR: reconstruction of the compatible connection and its Levi–Civita torsion-free branch;
3. GAP-10T-PALATINI (conditional): the minimal Cartan torsion equation is algebraic and invertible;
4. GAP-10L-CONN and GAP-10L-SYM (conditional): compatible transport and unique equivariant evolution preserve the Lorentz slice;
5. GAP-10I-SR and GAP-10I-PRESCRIBED: the affine flat representer and the exact augmented-holonomy criterion for prescribed curved coefficients;
6. GAP-10I-1S: the naive one-sided invertible curved route is a conditional no-go;
7. GAP-10D-PALATINI/UNIQUENESS (conditional): the minimal first-order branch and Lovelock hypotheses select Einstein–Λ;

8. GAP-10ψ-KIN/SYM: Lorentz-gauge flow or translation symmetry protects the classical metric under the stated hypotheses.

The undivided full-theory gaps are narrowed rather than fully closed:

1. GAP-10T-DYN: derive the selected first-order branch, spin current, normalization and possible extra torsion invariants from canonical UBT;
2. GAP-10I-CURVED: establish self-consistent on-shell selection, regularity and global continuation of (Θ, E, A, B, T);
3. GAP-10L-DYN: prove equivariance and well-posed uniqueness for the final complete dynamics;
4. GAP-10D: derive the low-energy Palatini/Lovelock assumptions, couplings and matter action from the fundamental theory;
5. GAP-10ψ: select the stability mechanism and exclude physical unstable non-gauge modes.

GAP-B-MASTER and GAP-U2Θ remain open. The earlier compact-ψ fiber closure remains a mathematically consistent historical candidate completion, but its weak selection and representer redundancy keep it outside the minimal canonical route.